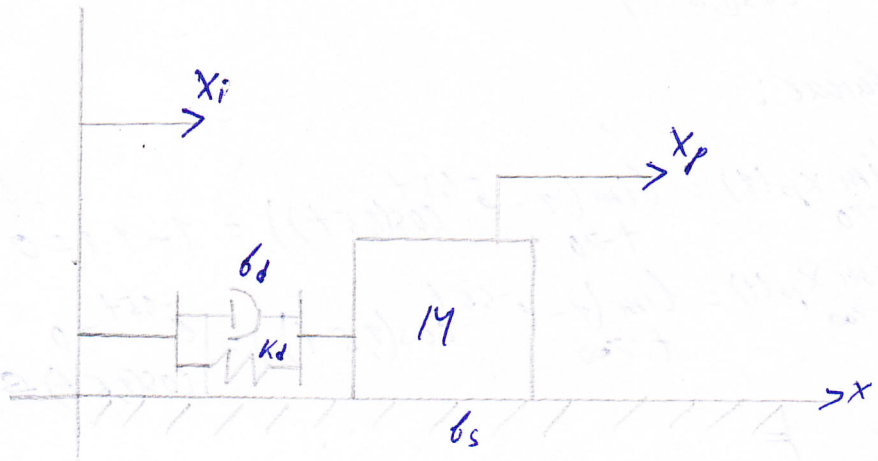
Дано:

$b_d = 1$

$k_d = 3$

$M = \frac{2}{3}$

$b_s = 1$



$G(s) = \frac{X_p(s)}{X_i(s)}$ - ?

$$\sum \bar{F} = m\bar{a}$$

Ox: $M\frac{dx_p^2}{dt} = -b_s\frac{dx_p}{dt} - b_d\frac{d(x_p - x_i)}{dt} - k_d(x_p - x_i)$

$$M\frac{dx_p^2}{d^2t} + b_s\frac{dx_p}{dt} + b_d\frac{dx_p}{dt} + k_d x_p = b_d x_i + k_d x_i$$

Н.У.: нулевые.

$$M X_p s^2 + b_s X_p s + b_d X_p s + k_d X_p = b_d X_i s + k_d X_i$$

$$X_p(Ms^2 + (b_s + b_d)s + k_d) = X_i(b_d s + k_d)$$

$$G(s) = \frac{X_p}{X_i} = \frac{b_d s + k_d}{Ms^2 + (b_s + b_d)s + k_d} = \frac{s+3}{\frac{2}{3}s^2 + (1+1)s + 3} = \frac{3s+9}{2s^2 + 6s + 9}$$

$$X_p(s) = G(s)\cdot X_i(s)$$

$x_i(t) = 1[t] \mapsto X_i(s) = \frac{1}{s}$

$$X_p(s) = \frac{1}{s}\cdot\frac{3s+9}{2s^2+6s+9} = \frac{3s+9}{s(2s^2+6s+9)} = \frac{A}{s} + \frac{Bs+C}{2s^2+6s+9} =$$

$$= \frac{2As^2 + Bs^2 + 6As + Cs + 9A}{s(2s^2+6s+9)}$$

$$\begin{cases} 2A + B = 0 \\ 6A + C = 3 \\ 9A = 9 \end{cases} \mapsto \begin{cases} A = 1 \\ B = -2 \\ C = -3 \end{cases}$$

$$X_p(s) = \frac{1}{s} + \frac{-2s-3}{2s^2+6s+9} = \frac{1}{s} - \frac{2s+3}{2s^2+6s+9} =$$

$$= \frac{1}{s} - \frac{2(s+1{,}5)}{2((s+1{,}5)^2 + 1{,}5^2)} = \frac{1}{s} - \frac{s+1{,}5}{(s+1{,}5)^2 + 1{,}5^2}$$

$$x_p(t) = 1 - e^{-1,5t}\cos(1,5t)$$

Моделирование:

$t \to 0$: $\lim\limits_{t\to 0} x_p(t) = \lim\limits_{t\to 0}(1 - e^{-1,5t}\cos(1,5t)) = 1 - 1\cdot 1 = 0$

$t \to \infty$: $\lim\limits_{t\to \infty} x_p(t) = \lim\limits_{t\to \infty}(1 - e^{-1,5t}\cos(1,5t)) = \left|\begin{matrix} e^{-1,5t} \to 0 \\ \cos(1,5t) \in [-1;1] \end{matrix}\right| =$

$= 1 - 0 = 1$

$t = 0$	$x_p(0) = 0$
$t = 1$	$x_p(1) = 0,98$
$t = 2$	$x_p(2) = 1,049$
$t = 3$	$x_p(3) = 1,00$

